

Syllabus

Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and Its Compounds, Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity, Light – Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current, Our Environment.

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. A patient is diagnosed with a condition where their glomerular filtration rate (GFR) is normal, but they are passing large amounts of dilute urine and frequently feel thirsty. Clinical tests show that their pancreas is functioning perfectly. Which of the following is the most likely physiological cause?
(A) Failure of the proximal convoluted tubule to reabsorb glucose.
(B) Hyposecretion of ADH (Antidiuretic Hormone) from the posterior pituitary.
(C) Hypersecretion of insulin leading to increased glucose in the filtrate.
(D) Over-activity of the loop of Henle in maintaining the osmotic gradient.
2. An object is moving with a constant speed of 3 m/s towards a convex mirror of focal length 1.5 m. At a particular instant, the object is 16 m away from the mirror. Find the instantaneous speed of the image at that instant.
(A) 0.42 m/s (B) 0.06 m/s
(C) 0.02 m/s (D) 0.2 m/s
3. 10^{-3} mole of KOH is added to 20 litres of pure water at 25°C. The pH will change by (assume no change in volume occurs)
(A) 2.7 (B) 4.1
(C) 7.3 (D) 11
4. A plant is placed in a sealed glass container with adequate water and light. After 24 hours, the concentration of CO_2 in the container remains exactly the same as it was at the start. What is the most logical explanation for this observation?
(A) The plant has stopped both photosynthesis and respiration.

- (B) The rate of photosynthesis is exactly equal to the rate of respiration (Compensation Point).
- (C) The plant is performing photosynthesis but not respiration.
- (D) The plant's stomata are completely closed, preventing any gas exchange.
5. The quantity of oxygen required for complete combustion of 0.5 mole of an organic compound $C_xH_yO_z$ is
- (A) $(x/2 + y/4 - z/2)$ moles
- (B) $(x/2 + y/8 - z/4)$ moles
- (C) $(x + y/8 - z/4)$ moles
- (D) $(x/4 + y/8 - z/2)$ moles
6. A fibre-optic cable of thickness 2.0 m is used to transmit light signals. A light ray enters the cable making an angle of 40° with the surface. The angle of refraction inside the cable is 28° . If the speed of light in air is 3×10^8 m/s, find the time taken by the light signal to pass through the cable.
- (A) 12.3 ns (B) 9.3 ns
- (C) 10.5 ns (D) 14.2 ns
7. Formula of a metallic oxide is M_4O_6 . Upon reduction with hydrogen the metallic oxide gives pure metal and water. 0.106 gm metal is produced by 8 mg of hydrogen after complete reduction. Atomic mass of the metal is
- (A) 18 (B) 260
- (C) 40 (D) 8
8. A specific plant hormone 'X' is applied to a fruit-bearing plant. It is observed that the fruits ripen quickly and the leaves start to fall off (abscission). Which hormone is 'X' and what is its primary nature?
- (A) Auxin; Growth-promoting.
- (B) Gibberellin; Growth-promoting.
- (C) Ethylene; Growth-inhibiting/Ripening agent.
- (D) Cytokinin; Cell division-promoting.
9. A proton is moving along the positive Y-axis with a uniform velocity. A uniform magnetic field is applied along the positive X-axis. What will be the effect on the motion of the proton?
- (A) The proton will experience a magnetic force along the positive Z-axis
- (B) The proton will experience a magnetic force along the negative Z-axis
- (C) The proton will not experience any magnetic force
- (D) The proton will continue to move in a straight line along the Y-axis
10. In the human male reproductive system, if the Vas Deferens is surgically blocked (Vasectomy), which of the following statements regarding the resulting semen is correct?
- (A) Semen will not be produced at all.
- (B) Semen will contain only secretions from the prostate and seminal vesicles but no sperm.
- (C) Semen will contain sperm, but they will be immobile.
- (D) Testosterone production will stop, leading to a lack of secondary sexual characters.

11. A hydrocarbon in LPG has the molecular formula C_3H_8 . Which of the following is true?
- (A) Its IUPAC name is Propene
(B) It is saturated
(C) It contains a double bond
(D) Its IUPAC name is Propanol
12. An optician prescribes spectacles of -3 D to a myopic person. This means the far point of that eye is approximately:
- (A) 43 cm (B) 3 m
(C) 30 cm (D) 0.33 m
13. A human male inherits his X-chromosome from which parent, and why does this determine the sex-linked inheritance pattern in males?
- (A) From his father; because the father provides the sex-determining chromosome.
(B) From his mother; because he must receive a Y-chromosome from his father to be male.
(C) From both parents; because X-chromosomes pair up during fertilization.
(D) From neither; X-chromosomes are formed de novo during embryonic development.
14. A sample of ascorbic acid (vitamin C) is synthesized in the laboratory; it contains 2.50g of carbon and 4.00g Oxygen. Another sample of Ascorbic acid isolated from citrus fruits contains 9.35g of carbon. How many grams of Oxygen does it contain? Which law do you assume in answering this question?
- (A) 7.25 : Law of conservation of mass
(B) 6.35: Law of definite proportions
(C) 8.47: Law of definite proportions
(D) 6.35: Law of multiple proportions
15. A photographer uses a convex camera lens to take a close-up picture of a flower placed 16 cm from the lens. The image formed on the camera sensor is magnified with $|m| > 1$. When the photographer moves the flower 8 cm closer to the lens, the camera settings are adjusted and the image again has the same magnification. The focal length of the camera lens is _____.
- (A) 12 cm (B) 14 cm
(C) 18 cm (D) 20 cm
16. A researcher observes that in a certain species of reptiles, the temperature at which the eggs are incubated determines whether the offspring will be male or female. This is an example of:
- (A) Genetic sex determination.
(B) Environmental sex determination.
(C) Acquired inheritance of traits.
(D) Natural selection for heat resistance.
17. Which of the following compounds is used as a flavoring agent in fruits and sweets?
- (A) Ethanoic acid (B) Propanal
(C) Ethyl ethanoate (D) Methanol
18. When bleaching powder ($CaOCl_2$) reacts with Conc HCl, chlorine gas is released. Which of the following is correct? (Chemical reactions and Equations)
- (A) $CaOCl_2 + 2 HCl \rightarrow CaCl_2 + Cl_2 + H_2O$

- (B) $\text{CaOCl}_2 + \text{HCl} \rightarrow \text{CaCl}_2 + \text{Cl}_2$
 (C) $\text{CaOCl}_2 + \text{HCl} \rightarrow \text{CaCl}_2 + \text{HClO}$
 (D) $\text{CaOCl}_2 + 2 \text{HCl} \rightarrow \text{CaCl}_2 + \text{Cl}_2 + \text{H}_2\text{O}$

19. Assertion (A): In mammals, the separation of the right and left sides of the heart is useful to keep oxygenated and deoxygenated blood from mixing.
Reason (R): Such separation allows a highly efficient supply of oxygen to the body, which is necessary for animals with high energy needs to maintain their body temperature.

- (A) Both A and R are true and R is the correct explanation of A.
 (B) Both A and R are true but R is NOT the correct explanation of A.
 (C) A is true but R is false.
 (D) A is false but R is true.

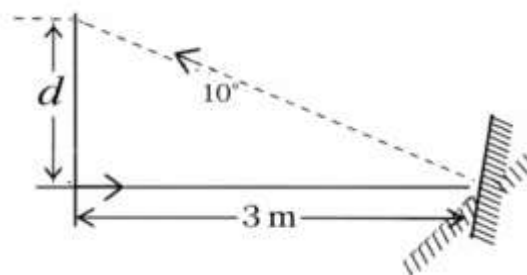
20. During a cold morning, the apparent sunrise is advanced more than usual. This suggests that

- (A) refractive index gradient of air has increased
 (B) scattering of blue light has increased
 (C) the Sun emits more red light
 (D) Earth's rotation has slightly changed

21. Copper does not react with water but reacts slowly with air in presence of moisture to form

- (A) Copper(I) oxide
 (B) Copper(II) oxide
 (C) Copper carbonate
 (D) Copper sulfate

22. A small plane mirror is mounted on a galvanometer coil and is initially perpendicular to a light beam incident normally on it as shown in the diagram. When an electric current is passed through the coil, it deflects the mirror by 10° . Calculate the horizontal displacement of the reflected spot of light on a screen located 3.0 m away.



- (A) 0.52 m (B) 0.87 m
 (C) 1.24 m (D) 1.09 m

23. Phenolphthalein is colourless in acidic medium but pink in basic medium. Which of the following shows this?

- (A) Vinegar (CH_3COOH) + phenolphthalein $\rightarrow$ Colourless
 (B) Lemon juice + phenolphthalein $\rightarrow$ Pink
 (C) NaOH solution + phenolphthalein $\rightarrow$ Colourless
 (D) Water + phenolphthalein $\rightarrow$ Pink

24. Assertion (A): Variation is useful for the survival of species over time, even if it is not necessarily beneficial for an individual.

Reason (R): Variations allow some individuals of a population to survive extreme environmental changes, such as a sudden rise in water temperature.

- (A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

25. A 1500 W electric room heater is designed to operate on a 240 V supply using a single heating coil. Due to a fault, an electrician cuts the same heating coil into four equal parts and connects all the parts in parallel across the same 240 V supply to restore heating. Ignoring temperature variation of resistance, what will be the new power consumption of the heater?

- (A) 1500 W (B) 3000 W
(C) 6000 W (D) 24,000 W

26. Which non-metal is essential for making fertilizers, matches, and gunpowder and dishwasher?

- (A) Sulfur (B) Phosphorus
(C) Oxygen (D) Nitrogen

27. If a plant shoot is illuminated with light coming only from the right side, it eventually bends toward the right. What is the physiological mechanism behind this "Phototropism"?

- (A) Auxin accumulates on the right (lighted) side, causing those cells to grow faster.
(B) Auxin is destroyed by light, so the plant stops growing on the right side.
(C) Auxin moves to the left (shaded) side, stimulating faster cell elongation there.
(D) The plant cells on the right side lose water and shrink, causing the stem to curve.

28. In a metro train, electromagnetic braking is used. When the metal disc attached to the wheel rotates between strong magnets, eddy currents are produced which oppose motion. If angular speed of the disc is doubled, the braking torque will:

- (A) remain same
(B) become 2 times
(C) become 4 times
(D) become 8 times

29. Which of the following metals does not react with cold water but reacts with steam to form a hydroxide and hydrogen gas?

- (A) Calcium (B) Magnesium
(C) Sodium (D) Potassium

30. A Mobile battery is rated at 11.1 V and has a capacity of 4400 mAh. How much total electrical energy is stored in the battery when it is fully charged? (electricity)

- (A) 48, 840 J (B) 175, 824 J
(C) 44, 400 J (D) 111,000 J

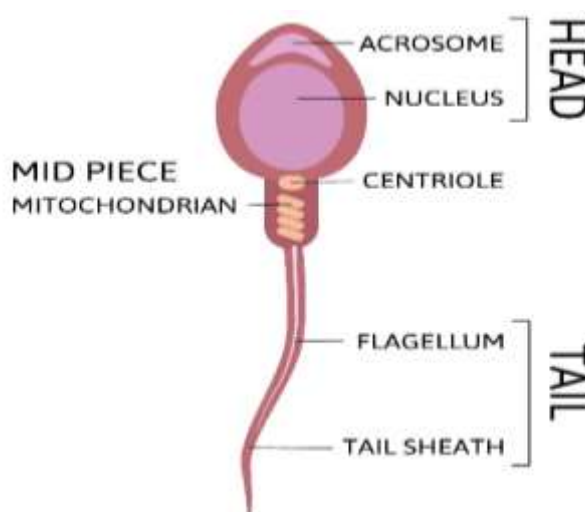
31. Consider a trait in a plant species where "Red Flowers" (R) are dominant over "White Flowers" (r). If a researcher crosses a heterozygous red-flowered plant with a white-flowered plant, what is the expected phenotypic ratio of the offspring?

- (A) 3 Red: 1 White
(B) 1 Red: 1 White
(C) All Red
(D) 1 Red: 2 Pink: 1 White

- 32.** Soaps are _____.
- (A) Calcium Salt of acids
 - (B) Magnesium salts of acids
 - (C) Sodium and potassium salts of long chain fatty acids
 - (D) Salts of bases
- 33.** A person experiences a sudden "Fight or Flight" situation. Their adrenaline levels spike. Which of the following describes the simultaneous physiological changes occurring in the digestive and circulatory systems?
- (A) Blood flow to the digestive system increases, and the heart rate decreases to conserve energy.
 - (B) Blood flow to the skeletal muscles increases, while blood flow to the digestive system and skin is reduced.
 - (C) The liver stops releasing glucose, and the heart rate stays constant to prevent high blood pressure.
 - (D) Both the stomach and the heart receive increased blood flow to manage the stress of the situation.
- 34.** The number of electrons flowing per second in the filament of a 100 W bulb operating at 240 V is : (Given $e = 1.6 \times 10^{-19} \text{ C}$)
- (A) 2.6×10^{18} electrons
 - (B) 2.9×10^{18} electrons
 - (C) 3.6×10^{16} electrons
 - (D) 1.6×10^{18} electrons
- 35.** Refining of impure copper with Zinc impurity is to be done by electrolysis using anode and cathode respectively as (Metals and non metals-ores)

- (A) pure Zinc, pure copper
- (B) pure Copper, pure Zinc
- (C) Impure Copper, pure Copper
- (D) impure Zinc, pure Zinc

- 36.** In a rare genetic case, a person is born with a non-functional 'Acrosome' in their sperm cells. Which specific step of the reproductive process will be primarily blocked?



- (A) The production of sperm in the seminiferous tubules.
 - (B) The motility of the sperm through the female reproductive tract.
 - (C) The penetration of the sperm through the protective layers of the egg.
 - (D) The fusion of the male and female nuclei after the sperm has entered the egg.
- 37.** For a thin convex lens, the radii of curvature of the two surfaces are 20 cm and 40 cm respectively. If the focal length of the lens is 25 cm, the refractive index of the material of the lens is (nearly)
- (A) 1.40
 - (B) 1.53
 - (C) 1.60
 - (D) 1.75

- 38.** Find the incorrect statement. (Carbon and its compounds)
- (A) 10% ethanol in gasoline is a good motor fuel.
- (B) Ethanol is a colourless liquid with characteristic of sweet odour and pure ethanol is called absolute alcohol
- (C) Orange Colour $Cr_2O_7^{2-}$ changes bluish green Cr^{3+} during the process of oxidation of ethanol and ethanoic acid
- (D) Denatured alcohol means 100% alcohol in the form of pure ethanol.
- 39.** A person has a blockage in their "Hepatic Portal Vein." This specific vessel carries blood from the small intestine directly to the liver. What will be the most immediate consequence after this person eats a meal?
- (A) The person will be unable to produce bile for digestion.
- (B) The liver will be unable to regulate the concentration of glucose entering the general circulation.
- (C) Deoxygenated blood will back up into the lungs, causing breathing difficulties.
- (D) The stomach will stop producing Hydrochloric acid (HCl).
- 40.** Which rule determines the direction of the magnetic field around a current-carrying conductor?
- (A) Right-Hand Thumb Rule
- (B) Left-Hand Rule
- (C) Fleming's Left-Hand Rule
- (D) Oersted's Rule
- 41.** A specific drug prevents the "re-uptake" of neurotransmitters from the synaptic cleft back into the pre-synaptic neuron. If this drug is applied to a neuromuscular junction, what would be the most likely physiological outcome?
- (A) The muscle will fail to contract because no signal is received.
- (B) The muscle will undergo prolonged or continuous contraction (tetany) due to the persistent presence of the neurotransmitter.
- (C) The electrical impulse will travel backward from the muscle to the brain.
- (D) The neurotransmitters will solidify in the synaptic cleft, blocking any further chemical movement.
- 42.** Which of the following is not an Oxidation reaction?
- (A) Bleaching of coloured objects using moist Chlorine.
- (B) Rancidity of fats.
- (C) Thermite process involving the reaction of Iron (III) oxide (or) Chromium (III) oxide, etc., with Aluminium.
- (D) The poling process involving the removal of impurities from a molten metal.
- 43.** During a period of prolonged starvation, the body first uses stored Glycogen, then begins to break down Fats, and finally starts breaking down Proteins. What is the physiological danger of the final stage (Protein breakdown)?
- (A) It leads to the overproduction of Vitamin C, which is toxic.

(B) It results in the loss of structural integrity of vital organs and muscles, including the heart.

(C) It causes the blood to become too alkaline, preventing oxygen transport.

(D) It prevents the liver from producing any more bile forever.

- 44.** In a mixer grinder, suppose one straight segment of the motor coil has a length of 0.25 m and carries a current of 1.2 A. If the magnetic field inside the motor is 0.6 T, what is the force experienced by this conductor?

(A) 0.18 N (B) 0.30 N

(C) 0.45 N (D) 0.60 N

- 45.** The reusable booster rockets of the U.S. space shuttle, uses a mixture of Aluminium and Ammonium perchlorate for fuel. A possible equation for this reaction is

$3\text{Al}_{(s)} + 3\text{NH}_4\text{ClO}_{4(s)} \rightarrow \text{Al}_2\text{O}_{3(s)} + \text{AlCl}_{3(s)} + 3\text{NO}_{(g)} + 6\text{H}_2\text{O}_{(g)}$ What mass of NH_4ClO_4 should be used in the fuel mixture for every kilogram of Al ?

(A) 3 kg (B) 3.388 kg

(C) 4.351 kg (D) 4 kg

- 46.** In a Mendelian cross, a researcher finds that the F₂ generation has a phenotypic ratio of 1:2:1 instead of the expected 3:1. What is the most likely biological explanation for this observation?

(A) The trait is controlled by multiple genes (Polygenic inheritance).

(B) The alleles for the trait show "Incomplete Dominance" or "Codominance."

(C) The recessive allele has undergone a mutation to become dominant.

(D) The environment has suppressed the expression of the dominant phenotype.

- 47.** A girl uses a concave mirror for makeup, mirror has focal length of 20 cm to see her face enlarged. She wants the image to be twice the size of her face. At what distance should she hold the mirror?

(A) 10 cm (B) 15 cm

(C) 20 cm (D) 30 cm

- 48.** During fog, visibility reduces because light of wavelength 600 nm is scattered by water droplets of size 2 μm . Which type of scattering mainly occurs?

(A) Rayleigh scattering

(B) Tyndall scattering

(C) No scattering

(D) Total internal reflection

- 49.** A researcher observes that a certain plant species has developed "Self-Incompatibility," a genetic mechanism that prevents its own pollen from fertilizing its own ovules. From an evolutionary perspective, what is the primary advantage of this complex trait?

(A) It ensures that the plant does not waste energy on producing seeds during a drought.

(B) It promotes "Outbreeding," thereby increasing genetic variation and the adaptability of the species.

(C) It allows the plant to reproduce asexually when pollinators are absent in the environment.

(D) It guarantees that the offspring will be exact clones of the parent plant, preserving successful traits.

50. On a hypothetical planet the major solvent is liquid Ammonia, not water. Ammonia auto ionises much like water ($2NH_3 \rightleftharpoons NH_4^+ + NH_2^-$). If instead of water, ammonia is used as a solvent, the acid base neutralisation reaction for the formation of NaCl is-

- (A) $NaNH_4 + NH_2Cl \rightarrow NaCl + 2NH_3$
 (B) $NaNH_2 + NH_4Cl \rightarrow NaCl + 2NH_3$
 (C) $NaNH_3 + NH_3Cl \rightarrow NaCl + 2NH_3$
 (D) $NaNH_4 + NH_4Cl \rightarrow NaCl + 2NH_4^+$

KEY

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|-------|-------|-------|-------|-------|-------|-------|
| 1) B | 2) C | 3) A | 4) B | 5) C | 6) A | 7) C |
| 8) C | 9) B | 10) B | 11) B | 12) D | 13) B | 14) C |
| 15) A | 16) B | 17) C | 18) A | 19) A | 20) A | 21) C |
| 22) D | 23) A | 24) A | 25) D | 26) B | 27) C | 28) B |
| 29) B | 30) B | 31) B | 32) C | 33) B | 34) A | 35) C |
| 36) C | 37) B | 38) D | 39) B | 40) A | 41) B | 42) D |
| 43) B | 44) A | 45) C | 46) B | 47) A | 48) B | 49) B |
| 50) B | | | | | | |